

Modbus_RTU Communication Description

Version: Version 1.0, November 2024

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The agreement number included in this agreement

- No. 3, No. 4, No. 5, No. 6

Communication data format

- During communication, data is sent back in the form of words (WORD - 2 bytes). In each word sent back, the high byte is in front and the low byte is in the back. If two words are sent back continuously (such as long integer), the high word is in front and the low word is in the back.

Data Type	Number of registers	Number of bytes	illustrate
Character	1	1	Send back two characters at a time, and fill them with 0 if they

Data Type	Number of registers	Number of bytes	illustrate
			are less than two.
Integer	1	2	Send back once, high byte first, low byte last
Long integer	2	4	Send back in two words, with the high word first and the low word last

Frame format

• Register content query (function code 03H)

The query's starting and ending addresses must be the starting and ending addresses of a complete data block; otherwise, the returned data will be incorrect. For example, if the device serial number register's starting address is 186 and its length is 12, the query's starting address cannot be between 186 and 198, and the ending address (starting address + number of registers read) cannot fall within this range.

Frame format sent by the host computer

Byte order	Code	Example	illustrate
0	Device Address	01H	Device address (1~247)
1	03H	03H	Function code

Byte order	Code	Example	illustrate
2	Starting register address high byte	00H	Register address high 8 bits
3	Starting register address low byte	10H	Register address lower 8 bits
4	Register number high byte	00H	High 8 bits of register number
5	Register number low byte	02H	Low 8 bits of register number
6	CRC16 check high byte	C0H	CRC16 check high 8 bits
7	CRC16 check low byte	CBH	CRC16 check lower 8 bits

The lower computer successfully parses and returns the frame format

Byte order	Code	illustrate
0	Device Address	Device address (1~247)
1	03H	Function code
2	Number of bytes of data sent back (N)	N = Number of registers requested*2
3	The first register data high byte	
4	The first register data low byte	
.....		
.....		

Byte order	Code	illustrate
	Nth register data high byte	
	Nth register data low byte	
N+3	CRC16 check high byte	All bytes before CRC must be checked
N+4	CRC16 check low byte	

Lower computer parses error and returns frame format

Byte order	Code	illustrate
0	Device Address	Device address (1~247)
1	03H	Function code
2	Number of bytes of data sent back (N)	N = Number of registers requested*2
3	The first 0	A total of N 0s are sent back
4	The second 0	
.....		
.....		
	The N-1th 0	
	Nth 0	

Byte order	Code	illustrate
N+3	CRC16 check high byte	All bytes before CRC must be checked
N+4	CRC16 check low byte	

Register read example:

Read the total mains current RMS value (starting register 202) to the total mains active power data, where the total mains current returns 5.0A, the mains frequency returns 50.0Hz, and the total mains active power returns 1200w

Host computer: 01 03 00 CA 00 03 25 F5

Lower computer: 01 03 06 00 32 13 88 04 B0 9F 6B

• Register content setting (function code 10H)

Frame format sent by the host computer

Byte order	Code	Example	illustrate
0	Device Address	01H	Device address (1~247)
1	10H	10H	Function code

Byte order	Code	Example	illustrate
2	Starting register address high byte	01H	Register address high 8 bits
3	Starting register address low byte	10H	Register address lower 8 bits
4	Register number high byte	00H	High 8 bits of the register number (always equal to 0)
5	Register number low byte	02H	Low 8 bits of register number
6	Number of bytes to be written (N)		$N = \text{number of registers} * 2$
7	The first register data high byte		
8	The first register data low byte		
.....			
.....			
	Nth register data high byte		
	Nth register data low byte		
N+7	CRC16 check high byte		CRC16 check high 8 bits
N+8	CRC16 check low byte		CRC16 check lower 8 bits

The lower computer successfully parses and returns the frame format

Byte order	Code	Example	illustrate
0	Device Address	01H	Device address (1~247)
1	10H	10H	Function code
2	Starting register address high byte	01H	Register address high 8 bits
3	Starting register address low byte	10H	Register address lower 8 bits
4	Register number high byte	00H	High 8 bits of the register number (always equal to 0)
5	Register number low byte	02H	Low 8 bits of register number
6	CRC16 check high byte	41H	CRC16 check high 8 bits
7	CRC16 check low byte	F1H	CRC16 check lower 8 bits

Lower computer parses error and returns frame format

Byte order	Code	illustrate
0	Device Address	Device address (1~247)
1	90H	Function code
2	Error Code	Error Code
3	CRC16 check high byte	CRC16 check high 8 bits

Byte order	Code	illustrate
4	CRC16 check low byte	CRC16 check lower 8 bits

Error code description

Code	illustrate
01H	A read-only register was operated
03H	Writing data that exceeds the acceptable range
07H	Registers that are not allowed to be modified in the current working mode

Register write example:

Set the output voltage (starting register 606) to 220V

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Equipment fault code		ULong	100	2	R	32-bit fault code, each bit corresponds to a fault code, see the fault code table for details , fault code 1 corresponds to bit 1, fault code 2 corresponds to bit 2, and so on	3 4 5 6
Get the warning code for unshielded processing			104	2		For details on the 32-bit warning code, see the warning code description.	3 4 5 6
Device Type		UInt	171	1	R		3 4 5 6
Protocol Number		UInt	184	1	R	0x03	3
Protocol Number		UInt	184	1	R	0x04	4
Protocol Number		UInt	184	1	R	0x05	5
Protocol Number		UInt	184	1	R	0x06	6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Device serial number		ASC	186	12	R		3 4 5 6
Power flow status		UInt	198	1	R	See the power flow flag description for details.	3 4 5 6
Working Mode		UInt	201	1	R	0: Power on mode 1: Standby mode 2: Mains mode 3: Off-grid mode 4: Bypass mode 5: Charging mode 6: Fault mode	3 4 5 6
Total mains current effective value	0.1A	Int	202	1	R		
Mains frequency	0.01Hz	Int	203	1	R		3 4 5 6
Total mains active power	1w	Int	204	1	R		

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Total city TV power	1VA	Int	205	1	R		
Mains charging power	1w	Int	206	1	R		3 4 5 6
Power Flow State Extension		UInt	207	1	R	See the power flow flag description for details.	3 4 5 6
Reserve			208	18			
Total inverter current effective value	0.1A	Int	226	1	R		
Inverter frequency	0.01Hz	Int	227	1	R		3 4 5 6
Total inverter active power	1w	Int	228	1	R		
Total inverter apparent power	1VA	Int	229	1	R		

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Inverter charging current	0.1A	Int	230	1	R		3 4 5 6
Inversion temperature	1°C	Int	231	1	R		3 4 5 6
Reserve			232	20			
Total output current effective value	0.1A	Int	252	1	R		4 6
Output frequency	0.01Hz	Int	253	1	R		3 4 5 6
Total output active power	1w	Int	254	1	R		4 6
Total output apparent power	1w	Int	255	1	R		4 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Total load percentage	1%	Int	256	1	R		4 6
Reserve			257	20			
Average battery voltage	0.1v	Int	277	1	R		3 4 5 6
Battery current filtered average value	0.1A	Int	278	1	R	Positive numbers indicate charging, negative numbers indicate discharging	3 4 5 6
Average battery power	1w	Int	279	1	R		3 4 5 6
Battery percentage	1%	UInt	280	1	R		3 4 5 6
DCDC temperature	1°C	Int	281	1	R		3 4 5 6
Reserve			282	20			

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Average total PV power	1w	Int	302	1	R		3 4
Average total PV charging power	1w	Int	303	1	R		3 4
Average total PV charging current	0.1A	Int	304	1	R		3 4
PV temperature	1°C	Int	305	1	R		3 4 5 6
Reserve			306	20			
Total external CT sampling power	1w	Long	326	2	R	Positive number: the grid outputs energy; negative number: energy is injected into the grid	
Reserve			328	10			

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Mains L1 voltage effective value	0.1v	Int	338	1	R		3 4 5 6
Mains L1 current effective value	0.1A	Int	339	1	R		
Mains L1 active power	1w	Int	340	1	R		3 4 5 6
Apparent power of mains L1	1VA	Int	341	1	R		
Inverter L1 voltage effective value	0.1v	Int	342	1	R		3 4 5 6
Inverter L1 current effective value	0.1A	Int	343	1	R		3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Inverter L1 active power	1w	Int	344	1	R	Positive numbers indicate inverter output, negative numbers indicate inverter input	3 4 5 6
Inverter L1 apparent power	1VA	Int	345	1	R		
Output L1 voltage effective value	0.1v	Int	346	1	R		3 4 5 6
Output L1 current effective value	0.1A	Int	347	1	R		3 4 5 6
Output L1 active power	1w	Int	348	1	R		3 4 5 6
Output L1 apparent power	1VA	Int	349	1	R		3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Output L1 load percentage	1%	Int	350	1	R		3 4 5 6
PV1 voltage average value	0.1v	Int	351	1	R		3 4 5 6
PV1 current average value	0.1A	Int	352	1	R		3 4 5 6
PV1 power average value	1w	Int	353	1	R		3 4 5 6
L1 external CT sampling power	1w	Long	354	2	R	Positive number: the grid outputs energy; negative number: energy is injected into the grid	
Reserve			356	20			
Mains L2 voltage effective value	0.1v	Int	376	1	R		

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Mains L2 current effective value	0.1A	Int	377	1	R		
Average power of mains L2	1w	Int	378	1	R		
Mains L2 apparent power	1VA	Int	379	1	R		
Inverter L2 voltage effective value	0.1v	Int	380	1	R		
Inverter L2 current effective value	0.1A	Int	381	1	R		
Average inverter L2 power	1w	Int	382	1	R	Positive numbers indicate inverter output, negative numbers indicate inverter input	

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Inverter L2 apparent power	1VA	Int	383	1	R		
Output L2 voltage effective value	0.1v	Int	384	1	R		4 6
Output L2 current effective value	0.1A	Int	385	1	R		4 6
Output L2 power average	1w	Int	386	1	R		4 6
Output L2 apparent power	1w	Int	387	1	R		4 6
Output L2 load percentage	1%	Int	388	1	R		4 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
PV2 voltage average value	0.1v	Int	389	1	R		3 4
PV2 current average value	0.1A	Int	390	1	R		3 4
PV2 power average value	1w	Int	391	1	R		3 4
L2 external CT sampling power	1w	Long	392	2	R	Positive number: the grid outputs energy; negative number: energy is injected into the grid	
Reserve			394	20			
Mains L3 voltage effective value	0.1v	Int	414	1	R		
Mains L3 current effective value	0.1A	Int	415	1	R		

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Average power of mains L3	1w	Int	416	1	R		
Mains L3 apparent power	1VA	Int	417	1	R		
Inverter L3 voltage effective value	0.1v	Int	418	1	R		
Inverter L3 current effective value	0.1A	Int	419	1	R		
Average inverter L3 power	1w	Int	420	1	R	Positive numbers indicate inverter output, negative numbers indicate inverter input	
Inverter L3 apparent power	1VA	Int	421	1	R		

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Output L3 voltage effective value	0.1v	Int	422	1	R		
Output L3 current effective value	0.1A	Int	423	1	R		
Output L3 power average	1w	Int	424	1	R		
Output L3 apparent power	1w	Int	425	1	R		
Output L3 load percentage	1%	Int	426	1	R		
PV3 voltage average value	0.1v	Int	427	1	R		
PV3 current average value	0.1A	Int	428	1	R		

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
PV3 power average value	1w	Int	429	1	R		
L3 external CT sampling power	1w	Long	430	2	R	Positive number: the grid outputs energy; negative number: energy is injected into the grid	
Reserve			432	168			
Output Mode		Uint	600	1	R/W	0: Single machine; 1: Parallel machine; 2: Three-phase combination-P1 3: Three-phase combination-P2 4: Three-phase combination-P3 5: Split-phase-P1 6: Split-phase-P2	3 4 5 6
(Primary) Output Priority		Uint	601	1	R/W	1: PV-mains-battery (SUB) [inversion priority]	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
						2: PV-battery-mains (SBU) 3: PV-mains-battery (SUF) [PV can be connected to the grid]	
(Secondary) Output Priority		Uint	602	1	R/W	0: Disable auxiliary output priority 1: PV-mains-battery (SUB) [priority for inversion] 2: PV-battery-mains (SBU) 3: PV-mains-battery (SUF) [PV can be connected to the grid]	3 4 5 6
Auxiliary output priority start time		Uint	603	1	R/W	Range: 0000~2359, thousands and hundreds are hours, tens and units are minutes	3 4 5 6
Secondary output priority end time		Uint	604	1	R/W	Range: 0000~2359, thousands and hundreds are hours, tens and units are minutes	3 4 5 6
Output priority for the current period		Uint	605	1	R	1: PV-mains-battery (SUB) [inversion priority] 2: PV-battery-mains (SBU)	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
						3: PV-mains-battery (SUF) [PV can be connected to the grid]	
Output voltage setting value	0.1v	Uint	606	1	R/W	2200: 220v output; 2300: 230v output; 2400: 240v output;	3 4 5 6
Output frequency setting value	0.01Hz	Uint	607	1	R/W	5000: 50Hz output; 6000: 60Hz output;	3 4 5 6
OP2 overload alarm setting	1%	Uint	608	1	R/W	Range: 10% ~ 100%	4 6
Is OP2 allowed to output?		Uint	609	1	R/W	0: Prohibit 1: Allow	4 6
OP2 output start time		Uint	610	1	R/W	Range: 0~23. Set the start time to 0 and the end time to 23, which means that OP2 can be output all day long if the conditions are met.	4 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
OP2 output end time		Uint	611	1	R/W	Range: 0~23. Set the start time to 0 and the end time to 23, which means that OP2 can be output all day long if the conditions are met.	4 6
Reserve			612	18			
Battery Type		Uint	630	1	R/W	0: AGM; 1: FLD; 2: USER; 4: Li2 6: Li4 8: LiB	3 4 5 6
Battery overvoltage protection point	0.1v	Uint	631	1	R/W		3 4 5 6
Battery charging priority		Uint	632	1	R/W	1: PV first (SOF) 2: PV and mains at the same level (SNU)	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
						3: Only PV charging allowed (OSO) 4: PV first to meet the output remaining charge (SOR)	
Secondary charging priority		Uint	633	1	R/W	0: Disable auxiliary charging priority 1: PV priority (SOF) 2: PV and mains at the same level (SNU) 3: Only PV charging is allowed (OSO) 4: PV priority satisfies the output remaining charge (SOR)	3 4 5 6
Auxiliary charging priority start time		Uint	634	1	R/W	Range: 0000~2359, thousands and hundreds are hours, tens and units are minutes	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Secondary charging priority end time		Uint	635	1	R/W	Range: 0000~2359, thousands and hundreds are hours, tens and units are minutes	3 4 5 6
Charging priority for the current period		Uint	636	1	R	1: PV first (SOF) 2: PV and mains at the same level (SNU) 3: Only PV charging allowed (OSO) 4: PV first to meet the output remaining charge (SOR)	3 4 5 6
Maximum charging voltage	0.1v	Uint	637	1	R/W		3 4 5 6
Float charge voltage	0.1v	Uint	638	1	R/W		3 4 5 6
Waiting time for constant	min	Uint	639	1	R/W	Range: 1~900min. If set to 0, the default is 10min.	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
voltage to float charge							
Maximum charging current	0.1A	Uint	640	1	R/W		3 4 5 6
Maximum AC charging current	0.1A	Uint	641	1	R/W		3 4 5 6
Maximum discharge current protection	1A	Uint	642	1	R/W	Range: 0 ~ 500A	3 4 5 6
Battery discharge recovery point in AC mode	0.1v	Uint	643	1	R/W	Set to 0 to indicate full recovery	3 4 5 6
Battery discharge	0.1v	Uint	644	1	R/W		3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
protection point in AC mode							
Off-grid mode OP1 battery low voltage protection point	0.1v	Uint	645	1	R/W		4 6
Off-grid mode battery discharge protection point	0.1v	Uint	646	1	R/W		3 4 5 6
Battery discharge SOC protection value in AC mode	1%	Uint	647	1	R/W		3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Battery discharge SOC recovery value in AC mode	1%	Uint	648	1	R/W		3 4 5 6
Off-grid mode OP1 battery discharge SOC protection value	1%	Uint	649	1	R/W		4 6
Off-grid mode battery discharge SOC protection value	1%	Uint	650	1	R/W		3 4 5 6
Battery Eq mode enabled		Uint	651	1	R/W	0: Disable; 1: Enable	3 4 5 6
Eq charging voltage	0.1v	Uint	652	1	R/W		3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
bat_eq_time	min	Uint	653	1	R/W	Range: 0 ~ 900	3 4 5 6
Eq timeout exit	min	Uint	654	1	R/W	Range: 0 ~ 900	3 4 5 6
The interval between two Eq charging	day	Uint	655	1	R/W	Range: 1 ~ 90	3 4 5 6
Force Eq Charge		Uint	656	1	W	1: Manually force Eq to charge once	3 4 5 6
Reserve			657	20			
Input Mode		Uint	677	1	R/W	0: APL; 1: UPS; 2: GNT	3 4 5 6
Buzzer mode		Uint	678	1	R/W	0: Mute in all situations; 1: Sound when the input source changes or when there is a specific warning or fault;	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
						2: Sound when there is a specific warning or fault; 3: Sound when there is a fault	
LCD backlight		Uint	679	1	R/W	0: Timed off; 1: Constant on	3 4 5 6
LCD automatically returns to the home page		Uint	680	1	R/W	0: No automatic return; 1: Automatic return after 1 minute	3 4 5 6
Energy saving mode switch		Uint	681	1	R/W	0: Energy saving mode off; 1: Energy saving mode on	3 4 5 6
Overload automatic restart		Uint	682	1	R/W	0: No restart in case of overload fault; 1: Automatic restart in case of overload fault	3 4 5 6
Over temperature		Uint	683	1	R/W	0: No restart on over-temperature fault;	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
automatic restart						1: Automatic restart on over-temperature fault	
Overload bypass enable		UInt	684	1	R/W	0: Disable; 1: Enable	3 4 5 6
Parallel PV detection mode		UInt	685	1	R/W	0: ; 1:	
External CT enable		UInt	686	1	R/W	0: Disable; 1: Enable	
Warning Mask [I]		ULong	687	2	R/W	Warnings corresponding to 1 are displayed normally, and warnings corresponding to 0 are blocked.	3 4 5 6
Dry Contact		UInt	689	1	R/W	0: Normal mode; 1: Grounded box mode;	3 4 5 6
Automatic mains output enable		UInt	690	1	R/W	0: No AC power output if the power button is not pressed 1: The AC power will	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
						automatically output if the power button is not pressed	
PV grid-connected maximum power	1w	Uint	691	1	R/W	Range: 200w ~ rated power	
Island detection enabled		Uint	692	1	R/W	0: Disable 1: Enable	
Boot method		Uint	693	1	R/W	0: Can be powered on locally or remotely; 1: Can only be powered on locally; 2: Can only be powered on remotely	3 4 5 6
Remote switch		Uint	694	1	R/W	0: Remote shutdown 1: Remote startup	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Exit fault lock state		Uint	695	1	W	1: Exit the fault lock state (only takes effect when the machine enters the fault mode)	3 4 5 6
Inverter Date		Uint	696	3	R/W	Three 16-digit numbers representing the year, month, and day (e.g., 07H E6H 00H 07H 00H 0EH represents July 14, 2022)	3 4 5 6
Inverter time		Uint	699	3	R/W	Three 16-digit numbers representing hours, minutes, and seconds (e.g., 00H 0DH 00H 0FH 00H 10H represents 13:15:16)	3 4 5 6
Power generation on that day	0.01kwh	Uint	702	1	R		3 4 5 6
Total power generation	0.01kwh	ULong	703	2	R		3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Zero power generation		Uint	705	1	W	0xAA: Clear all power generation data	3 4 5 6
Reset User Parameters			706	1	W	0xAA: User parameters are restored to default values (valid when working in non-off-grid mode)	3 4 5 6
Ground relay enable			707	1	R/W	0: Disable 1: Enable	4 6
Lithium battery activation time	min		709	1	R/W	Range: 6 ~ 300	3 4 5 6
Lithium battery automatic activation enable		Uint	858	1	R/W	0: Manual activation; 1: Automatic activation once after power on	3 4 5 6

Data Name	Unit	Data format	Starting address	Number of registers	Read and Write	illustrate	Valid protocol number
Lithium battery activation once		UInt	859	1	R/W	0: Invalid 1: Run once lithium battery activation	3 4 5 6

Host computer: 01 10 02 5E 00 01 02 08 98 8E 84

Lower computer: 01 10 02 5E 00 01 61 A3

Device register address

- R: Indicates readable, which means it supports 03H command. W: Indicates writable, which means it supports 10H command.
- Int: integer; Long: long integer; UInt: unsigned integer; ULong: unsigned long integer; ASC: ASCII code
- Max: take the maximum value; Min: take the minimum value
- At the same address, different protocol numbers have different data. Please refer to the column of valid protocol number. For example, at address 184, the device with protocol number 3 will return 3, and the device with protocol number 4 will return 4. The valid protocol number is empty and is considered invalid data.

- The addresses in the following table are expressed in decimal.

Fault code table

Fault Code	illustrate
bit0	Inverter overtemperature
bit1	PV overtemperature
bit2	DCDC over temperature
bit3	Battery overvoltage
bit4	Reserve
bit5	Output short circuit
bit6	Inverter overvoltage
bit7	Output overload
bit8	Busbar overvoltage
bit9	Bus soft start timeout
bit10	PV overcurrent
bit11	PV overvoltage

Fault Code	illustrate
bit12	DCDC overcurrent
bit13	Inverter overcurrent
bit14	Busbar low voltage
bit15	reserve
bit16	The inverter DC component is too high
bit17	PV2 current zero deviation is too large
bit18	Output current zero bias is too large
bit19	Inverter current zero bias is too large
bit20	Battery current zero bias is too large
bit21	PV1 current zero deviation is too large
bit22	Inverter low voltage
bit23	Inverter negative power protection
bit24	The host of the parallel system is lost
bit25	The parallel system synchronization signal is abnormal
bit26	Incompatible battery type
bit27	Incompatible parallel versions

Fault Code	illustrate
bit28	External CT setup repeat
bit29	Output 2 current zero deviation is too large

Warning Code Description

System warning is a 32-bit unsigned long integer. Each bit corresponds to a warning. Each bit can also be masked by the warning mask [I](#). After masking, the corresponding warning will not be read on the LCD or through commands.

• Warning code table

Warning Code	illustrate
bit 0	Mains zero-crossing loss
bit 1	Abnormal AC power waveform
bit 2	Mains overvoltage
bit 3	Mains undervoltage
bit 4	Mains overfrequency

Warning Code	illustrate
bit 5	Mains underfrequency
bit 6	PV1 undervoltage
bit 7	Overheating
bit 8	Low battery voltage
bit 9	Battery not connected
bit 10	Output overload
bit 11	Battery Eq Charging
bit 12	The battery is discharged to low voltage and has not yet been charged back to the recovery point
bit 13	Output power derating
bit 14	Fan stall
bit 15	PV energy is too low to be used
bit 16	Parallel communication interrupted
bit 17	Single and parallel output modes are inconsistent
bit 18	The voltage difference of the parallel batteries is too large
bit19	Lithium battery communication abnormality
bit20	The battery discharge current exceeds the set value

Warning Code	illustrate
bit 21	Island Detection
bit 22	PV2 undervoltage
bit23	Main output overload
bit24	Sec output load exceeds the set value
bit25	Sec output overload
bits 26 to 31	reserve

Power flow sign description

The power flow is represented by a 16-bit unsigned integer. The meaning of each bit is as follows:

- Power flow table (corresponding to register 198)**

Bit	illustrate
bit0~1	0: PV is not connected to the system

Bit	illustrate
	1: PV is connected to the system
bit2~3	0: Mains power is not connected to the system 1: Mains power is connected to the system
bit4~5	0: Battery neither charged nor discharged 1: Battery charging 2: Battery discharging
bit6~7	0: The system does not output load 1: The system outputs load
bit8	0: Not charging with AC power 1: Charging with AC power
bit9	0: PV not charging 1: PV charging
bit10	0: Battery icon on 1: Battery icon off
bit11	0: PV icon is on 1: PV icon is off
bit12	0: Mains icon is on 1: Mains icon is off
bit13	0: Load icon is on 1: Load icon is off

- **Power flow table expansion (corresponding to register 207)**

-

Bit	illustrate
bit0	0: PV1 is not connected to the system 1: PV1 is connected to the system
bit1	0: PV2 is not connected to the system 1: PV2 is connected to the system
bit2~3	0: AC input 1 is not connected to the system 1: AC input 1 is connected to the system
bit4~5	0: AC input 2 is not connected to the system 1: AC input 2 is connected to the system
bit6-7	0: AC input 3 is not connected to the system 1: AC input 3 is connected to the system
bit8	0: Main output with load 1: Main output without load
bit9	0: Sec output with load 1: Sec no output
bit10-15	Reserve

CRC checksum algorithm



Parameter model : CRC - 16 / MODBUS $X^{16} + X^{15} + X^2 + 1$

C language code

```
const char auchCRCHi [] = {
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
```

```
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1,
0x81, 0x40, 0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40
};
```

```
const char auchCRCLo [] = {
0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6, 0x06,
0x07, 0xC7, 0x05, 0xC5, 0xC4, 0x04, 0xCC, 0x0C, 0x0D, 0xCD,
0x0F, 0xCF, 0xCE, 0x0E, 0x0A, 0xCA, 0xCB, 0x0B, 0xC9, 0x09,
0x08, 0xC8, 0xD8, 0x18, 0x19, 0xD9, 0x1B, 0xDB, 0xDA, 0x1A,
0x1E, 0xDE, 0xDF, 0x1F, 0xDD, 0x1D, 0x1C, 0xDC, 0x14, 0xD4,
0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10, 0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3,
0xF2, 0x32, 0x36, 0xF6, 0xF7, 0x37, 0xF5, 0x35, 0x34, 0xF4,
0x3C, 0xFC, 0xFD, 0x3D, 0xFF, 0x3F, 0x3E, 0xFE, 0xFA, 0x3A,
0x3B, 0xFB, 0x39, 0xF9, 0xF8, 0x38, 0x28, 0xE8, 0xE9, 0x29,
0xEB, 0x2B, 0x2A, 0xEA, 0xEE, 0x2E, 0x2F, 0xEF, 0x2D, 0xED,
0xEC, 0x2C, 0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26,
0x22, 0xE2, 0xE3, 0x23, 0xE1, 0x21, 0x20, 0xE0, 0xA0, 0x60,
0x61, 0xA1, 0x63, 0xA3, 0xA2, 0x62, 0x66, 0xA6, 0xA7, 0x67,
0xA5, 0x65, 0x64, 0xA4, 0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F,
0x6E, 0xAE, 0xAA, 0x6A, 0x6B, 0xAB, 0x69, 0xA9, 0xA8, 0x68,
0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA, 0xBE, 0x7E,
0x7F, 0xBF, 0x7D, 0xBD, 0xBC, 0x7C, 0xB4, 0x74, 0x75, 0xB5,
0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73, 0xB1, 0x71,
0x70, 0xB0, 0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92,
0x96, 0x56, 0x57, 0x97, 0x55, 0x95, 0x94, 0x54, 0x9C, 0x5C,
0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B,
```

```

0x99, 0x59, 0x58, 0x98, 0x88, 0x48, 0x49, 0x89, 0x4B, 0x8B,
0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F, 0x8D, 0x4D, 0x4C, 0x8C,
0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42,
0x43, 0x83, 0x41, 0x81, 0x80, 0x40
};

unsigned short sModbusCrc16 ( INT8U * chMsg, INT16U dataLen )
{
    unsigned char ubCRCHi = 0xFF;
    unsigned char ubCRCLo = 0xFF;
    unsigned char duwIndex;
    while ( dataLen -- )
    {
        duwIndex = 0xff & ( ubCRCHi ^ * chMsg ++ );
        ubCRCHi = 0xff & ( ubCRCLo ^ auchCRCHi [ duwIndex ] );
        ubCRCLo = auchCRCLo [ duwIndex ];
    }
    return ( ubCRCHi << 8 | ubCRCLo );
}

```

--	--	--	--	--